MATH 201 Your name: ______________________

Learning target AD3, version 1

A wizard is developing a shrinking spell, but he needs to make sure it will work with his pointy hat, which is a cone whose height is three times its radius. Once he casts the spell, the height of his hat shrinks by 2 inches every second. How fast is the volume of his hat changing when the hat is just 3 inches tall? Give units on your answer.

Hints: Draw three pictures, label variables and constants, write down what you know and what you want to know, relate the variables, find the derivative wrt time, substitute and solve.

The volume of a cone is a third of the volume of the cylinder that contains it: $V = 1/3 \cdot \pi r^2 h$.

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Learning target AD3, version 2

Gravel is being dumped from a conveyor belt at a rate of 12 cubic feet per minute. The gravel forms a pile in the shape of a right circular cone whose base diameter and height are always the same. How fast is the height of the pile increasing when the pile is 12 ft high?